

VARSHA S

Senior Consultant | GIS Solutions Specialist | Qt C++ Developer

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PROFESSIONAL SUMMARY

Senior Consultant-profiled GIS and Qt C++ professional with 4+ years of experience delivering mission-critical desktop applications for defence-oriented geospatial and maritime situational awareness programmes. Skilled in Qt C++, QGIS, SQL, UI development, geospatial visualisation, application debugging, technical documentation, and end-to-end module delivery. Experienced in building GIS workflows involving SAR, LiDAR, radar, image analysis, map operations, map marking, database-driven utilities, and data purge applications. Known for translating client-specified requirements into reliable application modules, collaborating with cross-functional teams, and maintaining high-quality technical documentation across the software lifecycle.

PROFESSIONAL SNAPSHOT

Current Organisation	MicroGenesis TechSoft Pvt. Ltd.
Experience	4+ years in GIS application development, Qt C++, QGIS, SQL and database-driven desktop modules
Domain Exposure	Defence GIS, geospatial intelligence, SAR, LiDAR, IMSAS, maritime situational awareness, radar and image analysis
Role Positioning	Senior Consultant / GIS Application Specialist with focus on solution design, module ownership and delivery quality

CORE COMPETENCIES

Area	Strengths
GIS & Geospatial	GIS application workflows, QGIS, geospatial visualisation, spatial analysis, map operations, map marking, SAR, LiDAR, radar and image analysis
Software Engineering	Qt framework, C++, object-oriented programming, desktop application development, GUI design, module development, application debugging
Database & Data Management	SQL, database-driven modules, data purge utilities, data retrieval, data maintenance and operational support
Consulting & Delivery	Requirement analysis, client-specified module design, technical documentation, design documentation, testing, quality validation and team collaboration

PROFESSIONAL EXPERIENCE

MicroGenesis TechSoft Pvt. Ltd. | Senior Consultant / GIS Application Specialist | November 2021 - Present

- Contribute to GIS and QGIS module research, module design, technical introduction documentation and application design documentation using CASE tools.
- Design and develop client-specified user interfaces for GIS, SAR, LiDAR, radar, image analysis, map operation and map-marking workflows using Qt C++.
- Build resilient desktop application modules using C++, object-oriented programming principles and the Qt framework for geospatial visualisation and analysis.
- Develop and maintain SQL/database-driven modules supporting operational data handling, data purge workflows and application data management.
- Perform rigorous debugging and testing to improve application stability, reliability and functional quality across GIS desktop modules.
- Work as an on-site development partner supporting GIS projects involving SAR and LiDAR datasets, specialised defence workflows and application enhancement needs.
- Prepare and maintain technical documentation for developed modules, ensuring accurate project knowledge base and handover readiness.

PROJECT EXPERIENCE

Geographical Information System (GIS)

Client: Defence Research and Development Organisation (DRDO) | **Organisation:** MicroGenesis TechSoft Pvt. Ltd.

- Contributed to end-to-end development of a GIS-based desktop application covering complete application workflows.
- Designed and developed specialised UI modules for SAR, LiDAR, SAG Radar, image analysis, map operations and map marking.
- Used QGIS for GIS-based spatial analysis, geospatial processing and data visualisation activities.
- Supported development, testing and validation of modules to ensure reliable application behaviour for mission-critical GIS use cases.
- Prepared technical design documentation and module-level documentation to support maintainability and knowledge transfer.

Technologies: Qt C++, QGIS, SQL, GIS, SAR, LiDAR, Radar, Image Analysis

Indian Maritime Situational Awareness System (IMSAS)

Client: Defence Research and Development Organisation (DRDO) | **Organisation:** MicroGenesis TechSoft Pvt. Ltd.

- Enhanced existing application modules by improving GUI interfaces and user workflows.
- Designed and developed the Data Purge Desktop Application using Qt C++, SQL and database technologies.

- Performed comprehensive testing of developed modules and the overall application to ensure functional quality.
- Created and maintained technical documentation for all developed modules, including design and implementation details.
- Supported application improvements for maritime situational awareness workflows and data management requirements.

Technologies: Qt C++, SQL, Database Technologies, GIS, Desktop Application Development

TECHNICAL SKILLS

Programming	C++, SQL, Object-Oriented Programming concepts
Frameworks & Tools	Qt Framework, Qt Creator, QGIS, Visual Studio, CASE tools, Microsoft Office
Databases	SQL databases, database-driven desktop modules, data purge utilities
Operating Systems	RHEL, Red Hat 7.6, Ubuntu, Windows 10
Web Basics	HTML5, CSS3

EDUCATION

- Bachelor of Engineering (Electronics & Communication Engineering), Government College of Engineering, Bargur, Anna University - CGPA 7.8, 2019
- Diploma in Electronics & Communication Engineering, Adhiyamaan Polytechnic College, Hosur - 96%, 2016

CERTIFICATIONS & TRAINING

- Artificial Intelligence certification - Infosys Springboard
- YUVA - AI for All certification, authorised by Indian AI and offered via Coursera
- GIS certification - DRDO
- Core Java certification under PMKVY scheme - Rooman Technologies
- Skill development training in Computer Basics - Amazon Cares
- In-plant training at Uniq Technologies and workshop on Android App Development
- Market Research Internship - TESCOM Business Solutions